Net Meta Analysis

P&G/The Data Mine

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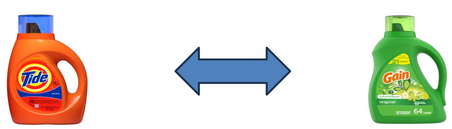
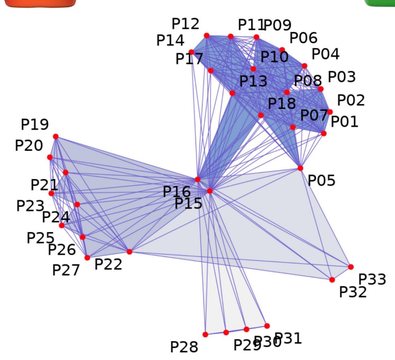
OVERVIEW

To test different versions of a products (i.e. laundry detergent), a study comparing every pair of products needs to be conducted. This is both time, effort, and money intensive. To overcome this, network meta-analysis is utilized to compare any pair of product, given that they have been compared to a baseline product. Using this technique, we developed an app, for P&G internal use, to automate this analysis for users who are not familiar with statistical software, saving even more time, effort, and money.

Statistical Background

Often, we have data from experiments directly comparing treatments to each other, but we want to estimate comparisons that don't exist experimentally. Network meta-analysis combines information from studies to provide these estimations.

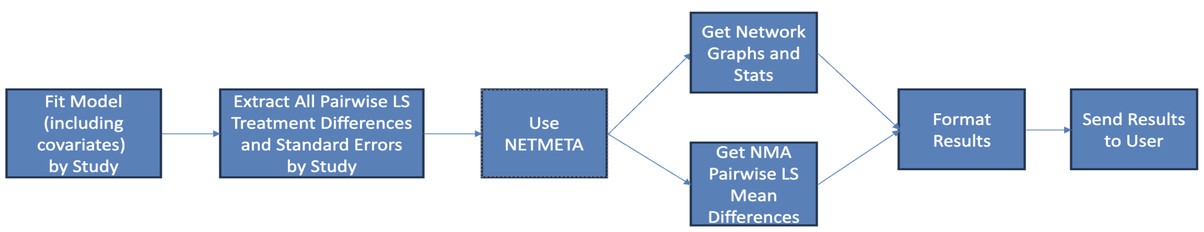


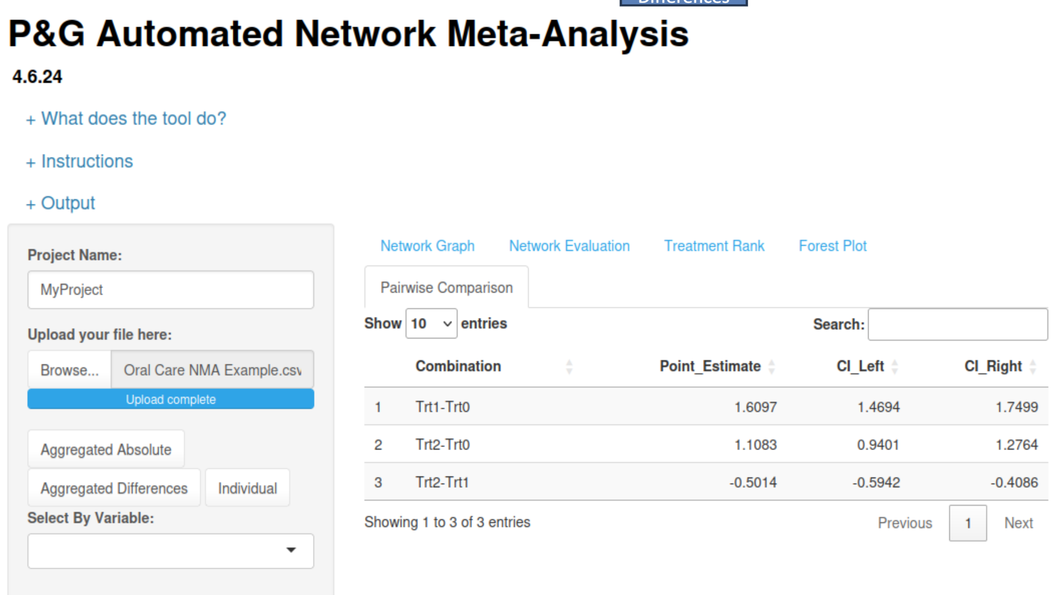


The above graph represents the pairwise comparisons that have been conducted between two products. Each vertex represents a product, and each edge represents a study.

R-Shiny Application

The purpose of this application is meant to allow everyone, even people without a statistical expertise, to be able to compare different studies, each with its own variability. The app takes in the data in the standard format – with studies and treatments. Then, it will run the entire net-meta analysis process by itself and generate the results and visualizations.





Results

Data Simulation

For testing purposes, a data simulation pipeline was developed in R, generating data as a 4d matrix. Since the data from P&G is sensitive, this acts as a placeholder to view, analyze, and run the net-meta-analysis on.

Future Plans

- **Exploring Jones et al. Methodology:** This is a more advanced method which accounts for covariates across studies. Currently, it's operational in SAS, but we're facing challenges with its application in R.
- **Commitment to Solutions:** We're dedicated to resolving these challenges. Our team will investigate further to ensure Jones Method can be effectively utilized in the future.

Conclusion

In our net-meta analysis implementation, we initially used the Jones methodology but faced setbacks due to limited support in R. We then transitioned to a combination of pre-processing and the net-meta library, enabling significant progress.

References

Balduzzi S, Rücker G, Nikolakopoulou A, Papakonstantinou T, Salanti G, Efthimiou O, Schwarzer G (2023). "netmeta: An R Package for Network Meta-Analysis Using Frequentist Methods." *Journal of Statistical Software*, **106**(2), 1–40. doi:10.18637/jss.v106.i02.